Answers to the Biology Practical Questions

a) (i) Common names of the given specimens

Specimen	Common Name
A	Housefly
B	Spider
C	Moss plant
D	Maize seedling
E	Bean seedling
F	Lizard
G	Leaf of pea
H	Goat pea
I	Fern
J	Cactus
K	Earthworm
L	Millipede
M	Centipede
N	Mushroom
O	Bee
P	Grasshopper
Q	Tilapia fish
R	Fern plant
S	Maize leaf
T	Pine tree branches
U	Hibiscus leaf

a) (ii) Four adaptations of specimen C (Moss plant) to its environment

1. **Rhizoids for anchorage**
   Moss has rhizoids that help attach it firmly to rocks, soil, or tree bark.
2. **Small size**
   Its small body reduces water loss and allows survival in moist habitats.
3. **Ability to absorb water directly**
   Moss absorbs water through the entire body surface because it lacks true roots.
4. **Production of spores**
   Spores are light and easily dispersed by wind, enabling reproduction in moist areas.

b) Class level of specimens

Specimen	Class
A (Housefly)	Insecta
B (Spider)	Arachnida

C (Moss)	Bryophyta
D (Maize seedling)	Monocotyledonae
E (Bean seedling)	Dicotyledonae
F (Lizard)	Reptilia
G (Pea leaf)	Dicotyledonae
H (Goat pea)	Dicotyledonae
I (Fern)	Pteridophyta
J (Cactus)	Dicotyledonae
K (Earthworm)	Oligochaeta
L (Millipede)	Diplopoda
M (Centipede)	Chilopoda
N (Mushroom)	Basidiomycetes/Fungi
O (Bee)	Insecta
P (Grasshopper)	Insecta
Q (Tilapia fish)	Pisces
R (Fern plant)	Pteridophyta
S (Maize leaf)	Monocotyledonae
T (Pine tree branches)	Gymnospermae
U (Hibiscus leaf)	Dicotyledonae

c) Two benefits of specimens D, E, K, N and T to human beings

Specimen D – Maize seedling

1. Source of food.
2. Used as livestock feed.

Specimen E – Bean seedling

1. Provides proteins as food.
2. Improves soil fertility through nitrogen fixation.

Specimen K – Earthworm

1. Aerates soil during burrowing.
2. Increases soil fertility through decomposition.

Specimen N – Mushroom

1. Used as food.
2. Decomposes dead organic matter.

Specimen T – Pine tree branches

1. Source of timber and paper pulp.

2. Helps in environmental conservation and rainfall formation.

d) Three reasons for placing specimen D and E into different classes

Maize Seedling (D)	Bean Seedling (E)
One cotyledon	Two cotyledons
Parallel venation	Reticulate venation
Fibrous roots	Tap roots

---

e) Difference between specimen B and E based on mode of nutrition

Specimen B (Spider)	Specimen E (Bean seedling)
Heterotrophic	Autotrophic
Feeds on other organisms	Manufactures own food by photosynthesis

f) Type of germination shown by specimen D and E

Specimen D (Maize seedling)

- Shows **hypogeal germination** because cotyledon remains below the soil.

Specimen E (Bean seedling)

- Shows **epigeal germination** because cotyledons emerge above the soil.

g) Economic importance of divisions

Specimen A – Arthropoda

1. Pollination by insects.
2. Source of food.
3. Biological control agents.
4. Some spread diseases.

Specimen H – Angiosperms

1. Source of food.
2. Source of medicine.
3. Provide timber.
4. Produce oxygen.

Specimen K – Annelida

1. Improve soil fertility.
2. Used in fishing bait.
3. Decomposition of organic matter.
4. Soil aeration.

Specimen N – Fungi

1. Used as food.
2. Used in medicine production.
3. Decompose dead materials.
4. Used in fermentation.

h) Four distinctive features of the class to which specimen M belongs (Centipede – Chilopoda)

1. Body divided into many segments.
2. One pair of legs per segment.
3. Possess poison claws.
4. Long flattened body.

i) Why scientists prefer binomial nomenclature instead of common names

1. Avoids confusion caused by local names.
2. Gives universal scientific names.
3. Shows relationship between organisms.
4. Each organism has only one accepted scientific name.

j) Differences between specimen K and L

Earthworm (K)	Millipede (L)
Soft body	Hard exoskeleton
No jointed legs	Many jointed legs
Body not divided into head and thorax	Distinct head and segmented trunk
Lives in moist soil	Lives in dark damp places

k) Six economic importance of specimen T (Pine tree)

1. Source of timber.

2. Used in paper production.
3. Produces resin and turpentine.
4. Prevents soil erosion.
5. Provides shade and windbreaks.
6. Helps in carbon dioxide absorption and oxygen production.

EXPERIMENT 03:

Demonstrating Gaseous Exchange Using a Mirror

Aim:

To demonstrate that carbon dioxide is a product of respiration (gaseous exchange) using a mirror.

Apparatus:

Mirror (clean and dry), human subject, stopwatch (optional), tissue for cleaning.

Procedure:

1. Breathe onto a clean, dry mirror.
2. Observe any changes on the surface of the mirror.

Q1: What is the purpose of using a mirror in this experiment?

A: The mirror is used to collect and observe the moisture and fog produced when a person exhales. This helps indicate the presence of water vapor and carbon dioxide from the lungs, demonstrating gaseous exchange.

Q2: What did you observe on the mirror after breathing on it?

A: A mist or fog formed on the mirror surface.

Q3: What does the mist on the mirror indicate?

A: The mist indicates the presence of water vapor in exhaled air, which is a product of respiration. It may also suggest that carbon dioxide is being exhaled, although CO_2 itself is invisible.

Q4: How does this experiment show gaseous exchange?

A: The experiment shows that exhaled air contains water vapor and carbon dioxide—both waste products of internal respiration. The fog on the mirror is due to condensation of water vapor, indirectly showing that gases are exchanged in the lungs.

Q5: Which gas is responsible for the cloudiness seen on the mirror?

A: Water vapor from exhaled breath condenses to form the cloudiness or mist seen on the mirror.

Q6: How could you confirm the presence of carbon dioxide in exhaled air more directly?

A: By passing exhaled air through limewater. If the limewater turns milky, it confirms the presence of carbon dioxide.

Q7: Why is it necessary to use a clean, dry mirror for this experiment?

A: To ensure that any condensation observed is solely due to exhaled breath and not from previous moisture on the mirror surface.

Q8: What biological process is being demonstrated in this experiment?

A: Respiration, specifically the gaseous exchange that occurs in the lungs.